

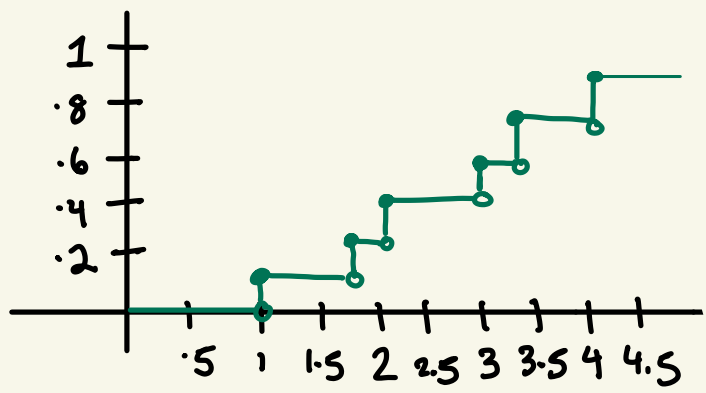
1.

data = [1, 1.75, 2, 3, 3.25, 4]

a)

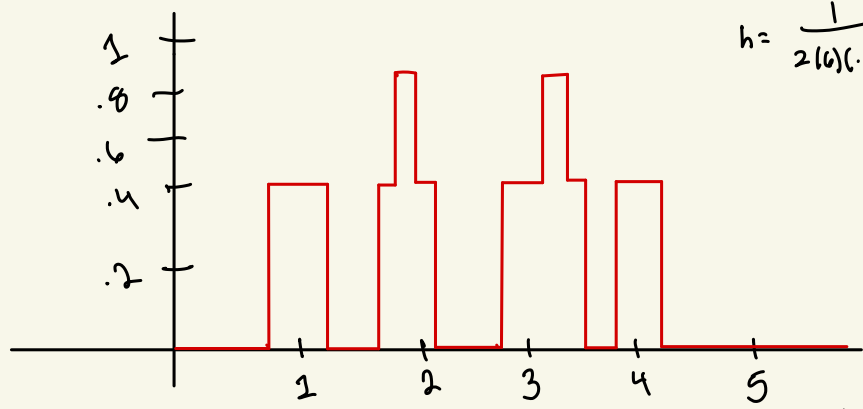
$$n=6$$

$F(x) = 0$	$x < 1$
$F(x) = 1/6$	$1 \leq x < 1.75$
$F(x) = 2/6$	$1.75 \leq x < 2$
$F(x) = 3/6$	$2 \leq x < 3$
$F(x) = 4/6$	$3 \leq x < 3.25$
$F(x) = 5/6$	$3.25 \leq x < 4$
$F(x) = 1$	$x \geq 4$



b)

$1 \rightarrow [0.8, 1.2]$	$3 \rightarrow [2.8, 3.2]$
$1.75 \rightarrow [1.55, 1.95]$	$3.25 \rightarrow [3.05, 3.45]$
$2 \rightarrow [1.8, 2.2]$	$4 \rightarrow [3.8, 4.2]$



$$h = \frac{1}{2(6)(.2)} = .4167$$

c)

$$h = 1.84 \cdot SD \cdot n^{(-0.2)}$$

$$\hookrightarrow \frac{1 + 1.75 + 2 + 3 + 4}{6} = 2.5$$

$$= 1.84 \times 1.1068 \times .6989 = 1.4232$$

	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	-1.5	2.25	
1.75	-.75	.5625	
2	-.5	.25	
3	.5	.25	
3.25	.75	.5625	
4	1.5	2.25	

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{6.125}{5} = 1.225$$

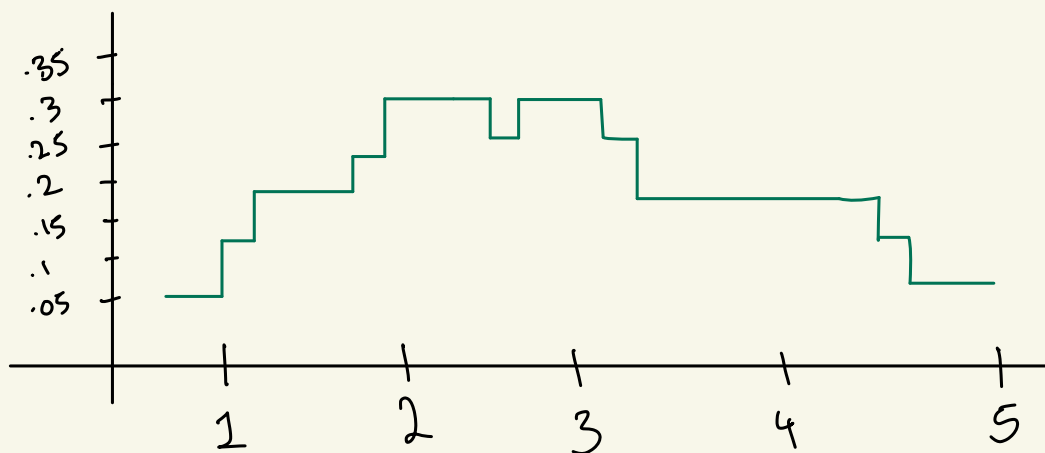
$$SD = \sqrt{s^2} = 1.1068$$

$$n^{-.2} = 6^{-.2} = .6989$$

d) $\frac{1}{2(6)(1.4232)} = .0585$

$1 \rightarrow [-.4232, 2.4232]$
 $1.75 \rightarrow [.3268, 3.1732]$
 $2 \rightarrow [.5768, 3.4232]$

$3 \rightarrow [1.5768, 4.4232]$
 $3.25 \rightarrow [1.8268, 4.6732]$
 $4 \rightarrow [2.5768, 5.4232]$



e) w) $h = 0.2$ all six points are visible and the spikes are easily told apart. w) Silverman bandwidth the boxes overlap a lot, so much so that the 6 points look merged, mostly a single-humped shape spanning almost all 6 points.

2) $x = [-1, -.5, 0, 0.5, 1]$ $n = 5$

a) $f(x) = \frac{1}{2hn} \sum_{i=1}^n 1(|x - x_i| \leq h)$

x_i	$ x_i - 0 $	< 0.6	Indicator
-1	1	no	0
-.5	0.5	yes	1
0	0	yes	1
0.5	0.5	yes	1
1	1	no	0
SUM			3

$f_{\text{hat}-h}(0) = \frac{3}{2(5)(0.6)} = \frac{3}{6} = 1/2$

b) $E[f_{\text{hat}}(0)] = \frac{(\phi(0+0.6) - \phi(0-0.6))}{(\phi(0.6) - \phi(-.6))} / (2)(.6)$

$= (0.72575 - 0.27425) / 1.2$

$= 0.3762$

c) $f(x) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(\frac{-x^2}{2}\right)$ $x=0$

$= \frac{1}{\sqrt{2\pi}} = 0.3989$

d) $E[f_{\text{hat}}(0)] \neq f(0)$

Not equal b/c there's a small neg. bias $(0.3762 - 0.3989)$
 b/c one's the density over the interval
 -0.6 to 0.6 and not @ a single point.

$$e) E[f_{\text{hext-h}}(0)] = \left(\frac{\phi(h) - \phi(-h)}{2h} \right) = \left(\frac{\phi(.1) - \phi(-.1)}{2(.1)} \right)$$

$$= \frac{.53983 - .46017}{.2} = 0.3983$$

h	$E[f_{\text{hext-h}}(0)]$	$f(0)$	Gap
0.6	0.3762	.3983	.0227
0.1	0.3983	.3983	.0007

Gap shrinks @ about 34 times when h drops from 0.6 to 0.1. Which matches our class notes where the expectation of h when it shrinks is $F'(0) = f(0)$.

f) Yes, KDE is biased but consistent. The bias goes to 0 only if h goes to 0 as n grows. In part c showed the bias shrinks as h shrinks. The Silverman rule sets h proportional to $n^{-1/5}$ so h goes to 0 as n grows. The bias then goes to 0 and the estimate approaches the true density.

g) Bias grows w/ h . $\uparrow h = \downarrow \text{Variance} \therefore \uparrow \text{bias}$; vice versa.

3) $x = 5, 7, 7, 8, 10$

$n = 5$

$h = 1.5$

a)
$$f_{\text{hist}}(x) = \frac{\text{Count}}{2nh} = \frac{\text{Count}}{2(5)(1.5)} = \frac{\text{Count}}{15}$$

x	Points within 1.5	Count	f_{hist}
6.0	5, 7, 7	3	.2
6.5	5, 7, 7, 8	4	.2667
7.0	7, 7, 8	3	.2
8.0	7, 7, 8	3	.2
8.5	7, 7, 8, 10	4	.2667

b)
$$f_{\text{hist}}(x) = \frac{1}{nh} \sum \left(\frac{1}{\sqrt{2\pi}} \right) e^{\left(-\frac{1}{2} \left(\frac{x_i - x}{h} \right)^2 \right)}$$

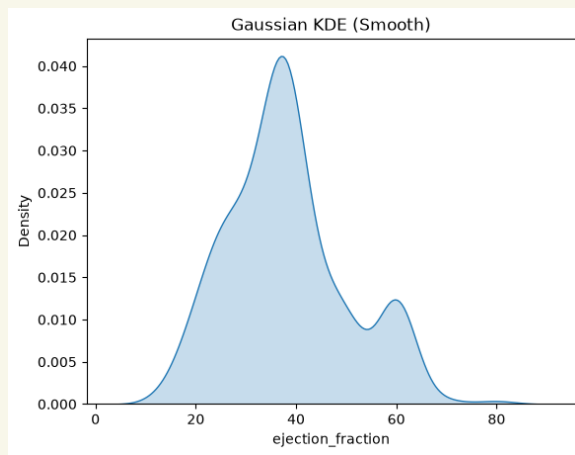
x	$f_{\text{hist}} (\text{Gaussian})$
6	0.1512
6.5	0.1687
7	0.1780
8	0.1674
8.5	0.1506

c) Uniform kernel is a step. w/ the Uniform kernel, each data point is either in or out. If it's within h of x , it counts as 1. If it's farther than h , it counts as 0. Nothing in between. So as x moves, the Count changes in whole steps, and the estimates jump. For example @ $x=6.5$ and $x=8.5$, where a data point is exactly 1.5 away. Gaussian kernel is a smooth. A point's weight shrinks gradually as it gets farther from x . NO Crossing point causes a sudden change in total, so the estimates changes smoothly.

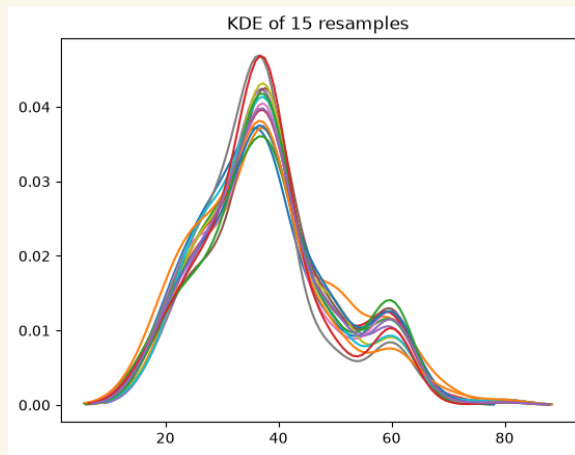
d) Gaussian kernel b/c it gives a smoother estimate. It weighs nearby points more than distant points. Uniform kernel is easier to compute by hand. It also ignores every point beyond h . That's also it's only advantage.

4)

a)



b)



c) The original plot is pretty reliable in its overall shape. When I run the resamples the peak is around the late 30's (like around 37), a shoulder around 20-30, and a secondary bump @ 60. All the resamples had the same pattern w/ tails being consistent b/c they sit near 0 in every curve. One thing amongst the resamples is a slight variations in heights (the main peak was diff per resample), left shoulder (differed slightly), a dip between the 2 peaks, and height @ 60. All of this happens b/c KDE @ x is an avg. of nearby observations, so it changes whenever a resample picks those observations $\uparrow$ or $\downarrow$. The peak shows the biggest change in h b/c the curve is the tallest there. The bump near 60 also changes b/c it comes from a cluster of 31 patients w/ an ejection fraction of 60, so a resample can pick that cluster $\uparrow$ or $\downarrow$.